

The permission value "0755" in octal notation corresponds to the binary representation "111 101 101" in terms of file permissions. These permissions are typically associated with Unix-like operating systems and determine who can perform various actions on the file. Let's break down what each digit in "0755" represents:

- The leftmost digit (0) indicates the special permissions. In this case, it's set to 0, indicating no special permissions.
- The middle digit (7) represents the owner's permissions. In binary, it's "111," which means the owner of the file has read (4), write (2), and execute (1) permissions.
- The rightmost digit (5) represents the group's permissions. In binary, it's "101," which means the group members have read (4) and execute (1) permissions but no write permission.
- The sequence "rwxr-xr-x" represents the permission string for this file.

So, the permission "0755" or "111 101 101" in binary means:

- The owner of the file can read, write, and execute it.
- Members of the group can read and execute it.
- Others (everyone else) can read and execute it but cannot write to it.

This permission setting is commonly used for executable files and scripts that should be readable and executable by all users but not writable to prevent unauthorized modifications.

In the context of Unix-like operating systems, file permissions are often represented using a set of numeric values where each digit corresponds to a different permission. The digits 4, 2, and 1 represent read, write, and execute permissions, respectively, for different categories of users (owner, group, and others).

When you see a combination like 8, 4, 2, and 1 used in this context, it typically indicates the numeric values associated with these permissions. Here's what they represent:

- 8 corresponds to the numeric value for "set user ID" (SUID) permission.
- 4 corresponds to the numeric value for "set group ID" (SGID) permission.
- 2 corresponds to the numeric value for the "sticky bit" permission.
- 1 corresponds to the numeric value for execute permission.